

Natural Coordinates for GPURec: Diagnosis and Practical Optimization Suggestions

A standalone mathematical and experimental assessment

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Abstract

The gene-wise duplication–loss–transfer model admits a useful hierarchical logit parameterization: one coordinate describes duplication/transfer intensity, one their relative mixture, and one loss relative to speciation. For fixed complete-history counts, including the appropriate survival-conditioning augmentation, its objective is a sum of three scalar binary likelihoods with an exactly diagonal Hessian. This does not diagonalize the observed likelihood: uncertainty over reconciliation histories reintroduces coupling and weak curvature. The most defensible practical strategy is therefore to use complete-history counts for a short bounded EM initialization and a cheap curvature seed, then learn the observed curvature with safeguarded per-family BFGS updates. Original log-rate bounds and convergence tests should remain authoritative. This note separates exact identities, implementation suggestions, and the experimental evidence available at the time of writing.

1 Diagnosis: the coordinates are only part of the difficulty

For one family, write

$$\theta = (\theta_D, \theta_L, \theta_T) = (\log_2 D, \log_2 L, \log_2 T), \quad Z = 1 + D + L + T.$$

Speciation is the reference event, with rate one, so

$$(p_S, p_D, p_L, p_T) = \frac{(1, D, L, T)}{Z}. \quad (1)$$

This discussion holds recipient and other non-rate parameters fixed, as in the gene-wise Coleman fits. Their contributions to each complete history are then independent of these three rates.

The existing log rates are already canonical exponential-family parameters, up to multiplication by $\ln 2$. Thus they are not an intrinsically inappropriate starting point. For fixed nonnegative expected event counts N_k , put $N = \sum_k N_k$. The complete-history negative log-likelihood surrogate in bits is

$$Q_\theta(\theta; N) = N \log_2 Z - N_D \theta_D - N_L \theta_L - N_T \theta_T + \text{constant}. \quad (2)$$

It is convex, with

$$\nabla Q_\theta = Np - (N_D, N_L, N_T)^\top, \quad C_\theta(\theta; N) = \ln(2)N\{\text{diag}(p) - pp^\top\}, \quad p = (p_D, p_L, p_T)^\top. \quad (3)$$

The observed family does not specify its event counts. Alternative histories can explain similar gene-tree patterns using different combinations of duplication, transfer, loss, and extinct lineages. Moving the rates changes that posterior mixture. Consequently, observed curvature can be much weaker than complete-data curvature, strongly coupled, and indefinite away from a solution. This is a statistical ambiguity as well as a numerical scaling problem.

2 An exact hierarchical coordinate system

2.1 Definition, interpretation, and inverse

Define $\psi = (u, v, w)$ by

$$\boxed{u = \log_2 \frac{D+T}{1+L}, \quad v = \log_2 \frac{T}{D}, \quad w = \log_2 L.} \quad (4)$$

Here u compares the event groups $\{D, T\}$ and $\{S, L\}$; v compares transfer with duplication within the first group; and w compares loss with speciation within the second. This directly exposes the observed duplication–transfer substitution rather than mixing it into two absolute rates.

Let $F(x) = \log_2(1 + 2^x)$. The inverse is smooth for positive rates:

$$\theta_D = u + F(w) - F(v), \quad \theta_L = w, \quad \theta_T = u + F(w) + v - F(v). \quad (5)$$

Equivalently, $D = 2^u(1 + 2^w)/(1 + 2^v)$, $L = 2^w$, and $T = 2^v D$. Evaluate F using a stable softplus implementation.

For $\sigma(x) = 1/(1 + e^{-x})$, set

$$a = \sigma(u \ln 2), \quad b = \sigma(v \ln 2), \quad c = \sigma(w \ln 2).$$

Then the four-way event choice factors into binary choices:

$$p_D = a(1 - b), \quad p_T = ab, \quad p_S = (1 - a)(1 - c), \quad p_L = (1 - a)c. \quad (6)$$

This is a computationally natural factorization, not a claim that the original log rates cease to be the canonical parameters of the model.

2.2 Exact separation of the fixed-count surrogate

Write $N_A = N_D + N_T$ and $N_B = N_S + N_L$, so $N = N_A + N_B$. Substituting (6) into the surrogate gives

$$Q_\psi(\psi; N) = \underbrace{NF(u) - N_A u}_{\text{group choice}} + \underbrace{N_A F(v) - N_T v}_{\text{transfer versus duplication}} + \underbrace{N_B F(w) - N_L w}_{\text{loss versus speciation}} + \text{constant}. \quad (7)$$

All counts in this differentiation are held fixed. In particular,

$$\nabla_\psi Q_\psi = \begin{pmatrix} Na - N_A \\ N_A b - N_T \\ N_B c - N_L \end{pmatrix}, \quad (8)$$

$$C_\psi(\psi; N) = \ln(2) \text{diag}(Na(1 - a), N_A b(1 - b), N_B c(1 - c)). \quad (9)$$

The Hessian is exactly diagonal. When all relevant counts are positive, the unconstrained minimizer is

$$u^+ = \log_2 \frac{N_A}{N_B}, \quad v^+ = \log_2 \frac{N_T}{N_D}, \quad w^+ = \log_2 \frac{N_L}{N_S}. \quad (10)$$

After inversion, these are precisely $D^+ = N_D/N_S$, $L^+ = N_L/N_S$, and $T^+ = N_T/N_S$. Thus an exact unconstrained EM update is the same operation in either coordinate system; a mere invertible reparameterization does not accelerate that exact update. Zero counts and the actual rate bounds require the constrained calculation below.

3 Survival conditioning and missing information

3.1 Use genuine augmented counts, not an arbitrary signed score decomposition

Let A_y be the unconditioned likelihood numerator, q the survival probability, and $e = 1 - q$ the extinction probability. The fitted likelihood is A_y/q . Since

$$\frac{A_y}{q} = A_y \sum_{m=0}^{\infty} e^m,$$

it admits an augmentation by the observed family and a geometrically distributed number of extinct “ghost” families. For this augmentation,

$$N_k = \mathbb{E}[n_k \mid y, \theta] + \frac{e}{q} \mathbb{E}[n_k \mid \text{extinct}, \theta] \geq 0. \quad (11)$$

These are the appropriate expected sufficient statistics for the EM surrogate. Data-minus-survival expressions can yield the same observed score, but score agreement alone does not establish an EM lower-bound argument for those signed quantities.

The existing adjoint calculation can expose the four independent event log-probability derivatives before the softmax chain rule:

$$a_k = \frac{\partial \ell}{\partial \log_2 p_k} = -N_k, \quad g_k = a_k - p_k \sum_j a_j = N p_k - N_k, \quad k \in \{D, L, T\}. \quad (12)$$

The extinction and survival-normalization derivatives must be included. Transfer recipient contributions are summed to obtain the family-level transfer count. This makes count extraction approximately a by-product of the ordinary reverse pass, not a second reconciliation solve.

3.2 What diagonal complete-data curvature does not imply

For an augmented complete history h , let $s_\psi(h) = \nabla_\psi \log_2 P(h, y \mid \psi)$. With posterior expectations evaluated at the current point, the observed Hessian in bits satisfies

$$H_\psi = I_{c,\psi} - I_{m,\psi}, \quad I_{c,\psi} = C_\psi(\psi; \mathbb{E}[n \mid y, \psi]), \quad I_{m,\psi} = \ln(2) \text{Cov}[s_\psi(h) \mid y, \psi]. \quad (13)$$

The covariance generally has off-diagonal entries. Diagonalizing the first term therefore neither diagonalizes the observed Hessian nor removes weakly identified directions. A diagonal complete-data metric can also be much too stiff: the subtraction in (13) can be substantial.

For clarity, a nonlinear Hessian transformation is not just a change of basis. If $J = \partial \theta / \partial \psi$ and $g = \nabla_\theta \ell$, then

$$H_\psi = J^\top H_\theta J + \sum_k g_k \nabla_\psi^2 \theta_k. \quad (14)$$

For (5), the Jacobian and correction are

$$J = \begin{pmatrix} 1 & -b & c \\ 0 & 0 & 1 \\ 1 & 1-b & c \end{pmatrix}, \quad (15)$$

$$\sum_k g_k \nabla_\psi^2 \theta_k = \ln(2)(g_D + g_T) \text{diag}(0, -b(1-b), c(1-c)). \quad (16)$$

This gradient-dependent term explains why the hierarchy can remove much apparent negative curvature away from an optimum. At an interior stationary point it vanishes, and Hessian inertia is preserved under the remaining congruence.

At a regular interior EM fixed point, the local EM convergence matrix is $R = I_c^{-1} I_m$. Its eigenvalues, equivalently those of $I_c^{-1/2} I_m I_c^{-1/2}$, are unchanged by a smooth invertible reparameterization of the same exact EM map. A coordinate change can improve numerical scaling, but cannot remove the underlying missing-information fraction. These stationary-point statements should not be applied to an arbitrary off-stationary Hessian. At a constrained optimum the relevant second-order analysis uses feasible directions and the constraint geometry.

4 Bounds are a substantive part of the problem

Coleman uses $r_{\min} = 10^{-6}$ and $r_{\max} = 2$. Let $\ell_b = \log_2 r_{\min}$ and $h_b = \log_2 r_{\max}$. The original box becomes

$$\ell_b \leq w \leq h_b, \quad (17)$$

$$\ell_b \leq u + F(w) - F(v) \leq h_b, \quad (18)$$

$$\ell_b \leq u + F(w) + v - F(v) \leq h_b. \quad (19)$$

Replacing this region by three independent coordinate intervals changes the optimization problem. Mapping a proposal back and clipping its rates preserves feasibility, but changes the actual step; predicted decrease and acceptance must be evaluated consistently with that step.

The bounded EM M-step is particularly cheap in the original log rates. For each assignment of a coordinate to its lower bound, free state, or upper bound, let B be the pinned coordinates and b_j their log-rate bounds. Set

$$c_B = 1 + \sum_{j \in B} 2^{b_j}, \quad A_B = N_S + \sum_{j \in B} N_j.$$

The free stationary solution is

$$\theta_k^+ = \log_2 \frac{N_k c_B}{A_B}, \quad k \notin B. \quad (20)$$

Enumerate the $3^3 = 27$ active-set assignments and check feasibility and KKT signs. For the maximizing score $s_k = N_k - N p_k$, the signs are $s_k \leq 0$ at a lower bound, $s_k = 0$ when free, and $s_k \geq 0$ at an upper bound. This avoids irreversible-pinning mistakes: pinning one rate can require releasing another. Positive-total counts and finite bounds make the surrogate strictly convex in θ , so a feasible KKT solution is its unique minimizer.

The convergence criterion should remain the original projected log-rate gradient, $\|P_{\theta} g_{\theta}\|_{\infty} < 10^{-3}$. Applying the same numerical threshold to a rescaled-coordinate gradient would silently change the test.

5 Recommended optimization strategy

5.1 Short EM initialization, followed by observed-curvature learning

The strongest currently supported combination is:

1. Start from the same feasible rates as the baseline. Take two or three bounded EM steps using counts extracted in the existing gradient passes. Reuse the already-built model and parsed families.

2. At the final warm-up endpoint θ_K , form $C = C_\theta(\theta_K; N^{K-1})$ using the latest fixed count set and (3). This is the exact Hessian of that fixed-count surrogate, not the observed Hessian at θ_K .
3. Calibrate its scale using the already-paid latest observed-gradient secant, $s = \theta_{K-1} - \theta_{K-2}$ and $y = g_{K-1} - g_{K-2}$. When both quadratic quantities are finite and positive, use

$$\alpha = \frac{s^\top y}{s^\top C s}, \quad B_0 = \alpha C.$$

Otherwise retain a safe positive scale. Apply a safeguarded BFGS correction on coordinates eligible at both secant endpoints.

4. Continue with the original per-family trust-region/BFGS method, scheduled exact curvature refreshes, bound handling, and convergence tests. Keep the full 3×3 matrix: its algebra is cheap compared with a reconciliation gradient.

For a positive-definite B and suitable positive-curvature secant, the familiar update is

$$B^+ = B - \frac{B s s^\top B}{s^\top B s} + \frac{y y^\top}{s^\top y}.$$

Practical curvature floors and free-coordinate safeguards remain necessary. The calibration learns only one direction and uses a secant preceding the endpoint. It is a useful approximation, not a proof that B_0 equals observed information or globally majorizes the objective.

Exact posterior counts and an exact bounded M-step imply EM likelihood monotonicity. Pruned FP32 adjoints, finite fixed-point tolerances, and rounding the endpoint back to model precision weaken that software guarantee. A double output buffer does not turn FP32-derived counts into exact counts. Observed likelihood checks and the unchanged final certificate remain important.

5.2 Interpret adaptive curvature as local coordinates

For a safeguarded positive-definite approximation B_f in one family’s free subspace, write $B_f = R_f^\top R_f$. Then

$$z_f = R_f(\theta_f - \bar{\theta}_f) \tag{21}$$

gives an identity quadratic curvature in the local model. These are family-specific, adaptive coordinates: they can scale and rotate to follow the posterior ambiguity. They need not be materialized in the likelihood kernels; the step can be solved directly in θ , with its simple box constraints.

A crucial distinction is that an SPD preconditioning metric transforms as a tensor, while an ordinary off-stationary Hessian has the extra term in (14). Pulling the diagonal matrix (9) back without that term defines a possible metric, but is not generally the exact fixed-count Hessian in log rates. Computing (3) directly avoids this ambiguity.

Whitening alone is not an additional acceleration over solving the same quadratic model exactly in the original variables. It helps when it improves the approximation, numerical conditioning, or trust-region metric. Any changed trust-region geometry should therefore be tested as its own algorithmic choice.

5.3 Priorities for further experiments, not established gains

1. Prefer count-informed full-matrix curvature over a permanently diagonal hierarchical optimizer. The latter discards precisely the posterior coupling responsible for difficult families.

2. Test any new scaling or metric with matched bounds, acceptance rules, and curvature-refresh schedules. A negative result for one hierarchical replay is not a theorem that every hierarchical hybrid fails.
3. Consider improving the latent augmentation only if missing information remains the bottleneck. Collapsing avoidable latent variables or using a valid alternative augmentation can change EM’s convergence spectrum; renaming the same parameters cannot. Such changes require a new likelihood/EM derivation, not ad hoc subtraction of ghost counts.
4. Do not assume that a better median condition number or a shorter path to saved optima lowers runtime. Charge additional gradient evaluations, Hessian probes, rejected trials, initialization, and difficult-family completion. Saved fitted optima are evaluation references, never inputs to a proposed initialization.

6 Experimental evidence and its limits

6.1 Hierarchical coordinates improve geometry, but have not won the runtime comparison

On 500 saved Coleman-family states, the hierarchical formulas, Jacobian, and Hessian transformation were checked numerically against differentiation. At the post-Adam point, observed-Hessian indefiniteness fell from 64% in log rates to 9% in the hierarchy. The median relative curvature drift to the fitted endpoint was still about 1.06.

In the subsequent 200-family GPU replay, a scaled hierarchical diagonal reached 186/200 fresh certificates after the longer bounded run, with NLL about 1.37 bits worse than the fully converged production baseline. Some failures were at coupled rate-bound corners. That replay lacked the production exact refresh at step 15, so it was not an otherwise-identical full hybrid comparison. The evidence supports rejecting that tested configuration, not ruling out every coordinate-aware method.

6.2 Count-informed initialization has measured gains

Table 1 gives the matched 500-family RTX 4090 prototype runs recorded in the campaign. Both EM prototypes include their redundant warm-model construction in wall time. All three use the same production freeze-time projected-gradient criterion and certify all 500 families.

Initialization	Wall time (s)	Gradient clade equivalents	NLL (bits)
Three Adam steps	108.33	17.273	1618461.966967
Two EM steps + scaled curvature	91.51	13.660	1618461.994534
Three EM steps + scaled curvature	87.21	13.299	1618461.952767

Table 1: Matched local prototype evidence; these are not full-Coleman H100 timings.

The three-EM-step prototype is approximately 19.5% faster than the Adam prototype. Small aggregate NLL differences do not imply identical family-wise solutions: four families change NLL by more than 0.01 bits relative to Adam, with both improvements and regressions. The recorded integrated three-step gate subsequently took 81.42 s on 500 families and certified all of them; that number should not be presented as a matched integrated Adam speedup.

The first locally recorded full-Coleman H100 result is also encouraging: two EM steps followed by the existing optimizer completed all 5,124 families in 396.31 s, with 28 Newton steps and all 5,124

freeze-time certificates. This is about 23.9% below the historical 520.5 s record. Its fit-reported NLL is 9049360.7436 bits. However, the available contemporaneous Adam run took 634.02 s with unusually expensive startup/warm-up; it is not a fair sole denominator for a warm speedup claim. The warmed paired control, repetitions, and fresh common-model likelihood/gradient audits were not yet available in the local result snapshot used here. The full-scale comparison is therefore preliminary, not a completed paired validation.

For fair accounting, define gradient clade equivalents by

$$W_g = \sum_j \frac{\text{clades resident in the model evaluated on gradient call } j}{\text{clades in the initial full population}}.$$

Frozen families still cost work until a replan removes them. Report Hessian cost separately; its probes are not automatically equivalent to a fixed number of gradient calls. The campaign’s certificate uses the pruned FP32 gradient recorded at freezing. It is not an unpruned mathematical stationarity proof.

7 Conclusion

There is a natural and exact hierarchical factorization of the event model. It makes the fixed-count problem separable and provides useful geometric diagnostics. The hard part of the observed problem, however, is the changing posterior over histories, together with active bounds and expensive gradient passes. A globally fixed coordinate transformation cannot erase that ambiguity.

Use complete-history structure to enter a good region cheaply and initialize curvature; use observed-gradient information to learn each family’s actual local geometry; preserve the original constrained problem and its certificate.

Provenance and scope

This is an independent side-conversation note. It does not modify the earlier mathematical report, the optimization code, or the running campaign. The evidence is a snapshot of the local files read on 6 September 2026, not a continuously updated campaign status. In particular, the older report’s statement that EM had not been measured is superseded by the later experiment records.

The model and earlier coordinate diagnosis are documented in `agent_mathematical_report.tex`. Hierarchical identities, numerical checks, and replay limitations are recorded in `experiments/coleman_sol_20260906/geometry/PROGRESS.md`. The survival-count and bounded-M-step review is in `experiments/coleman_sol_20260906/geometry/EM_MATH_REVIEW.md`. Local timing and likelihood comparisons are in `experiments/coleman_sol_20260906/em/PROGRESS.md`. The preliminary full-H100 result is `experiments/coleman_sol_20260906/results/em_full_a.json`, with its cold control in `experiments/coleman_sol_20260906/results/baseline_interactive_a.json`. Implementation details were checked against `gpurec/fit/em_warmup.py` and the associated count API discussion.